HTML5

A Comprehensive Introduction

From Networks to Web Pages

Topics Covered

- Web Ecosystems & Protocols
- HTML5 History & Syntax
- Core Tags with Full Examples
- Forms, Tables, Media & More
- Accessibility Best Practices

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Table of Contents

Table of Contents 2
1. Introduction: The Web Ecosystem 3
1.1 Networks 3
1.2 The Internet vs the World Wide Web 3
1.3 Protocols 3
1.4 DNS — Domain Name System 3
1.5 Web Browsers 3
1.6 Web Servers 4
1.7 Client–Server Model 4
1.8 URL Anatomy 4
2. What Is HTML5? 5
2.1 A Brief History 5
2.2 The Web Trinity 5
2.3 Key Improvements over HTML4 5
3. HTML5 Document Syntax 6
3.1 The Boilerplate 6
3.2 Line-by-Line Explanation 6
3.3 Tag Syntax Rules 6
4. Core HTML5 Tags 7
4.1 Document Structure Tags 7
4.2 Semantic Layout Tags (HTML5) 7
4.3 Heading Tags 8
4.4 Text Content Tags 8
4.5 List Tags 10
4.6 Link Tags 10
4.7 Image & Media Tags 11
4.8 Table Tags 12
4.9 Form Tags 12
4.10 Miscellaneous & HTML5-Specific Tags 15
5. Global Attributes 16
6. HTML5 Best Practices & Accessibility 17
6.1 Structural Best Practices 17
6.2 Accessibility (a11y) 17
6.3 Validation & Performance 17
7. Quick Reference: HTML5 Tag Summary 18

1. Introduction: The Web Ecosystem

Before writing a single HTML tag, it is essential to understand the infrastructure that makes web pages possible — from physical cables to the browser rendering your screen.

1.1 Networks

A network is a collection of interconnected devices that can communicate and share resources. Networks are classified by scale:

- LAN (Local Area Network) — devices in one location (e.g., office Wi-Fi)
- WAN (Wide Area Network) — spans cities, countries, or continents
- The Internet — the global WAN connecting billions of devices worldwide

1.2 The Internet vs the World Wide Web

Concept	Definition
The Internet	The global physical/logical network infrastructure (cables, routers, servers, protocols)
The World Wide Web	A service that runs on the Internet; a system of hyperlinked HTML documents accessed via HTTP
Key distinction	The Internet is the road; the Web is one type of traffic on that road (email, FTP, etc. are others)

1.3 Protocols

A protocol is a set of agreed-upon rules governing how data is transmitted between devices.

Protocol	Purpose
TCP/IP	Foundational suite: TCP ensures reliable delivery; IP handles addressing and routing
HTTP	HyperText Transfer Protocol — transfers web pages (plain text)
HTTPS	HTTP Secure — HTTP encrypted via TLS/SSL; required for login forms and payments
FTP / SFTP	File Transfer Protocol — uploads/downloads files between server and client
SMTP / IMAP	Email protocols — SMTP sends; IMAP retrieves

1.4 DNS — Domain Name System

DNS is the Internet's phone book. Humans use domain names (www.example.com); computers use IP addresses (93.184.216.34). DNS translates one to the other automatically when you type a URL.

1.5 Web Browsers

A browser is a software application that requests HTML documents from servers and renders them visually. Major browsers and their rendering engines:

- Chrome / Edge — Blink engine
- Firefox — Gecko engine
- Safari — WebKit engine

1.6 Web Servers

A web server stores files and responds to HTTP requests. Popular servers include Apache (open-source, cross-platform) and Nginx (high-performance, often used as a reverse proxy).

1.7 Client–Server Model

The web operates on a request–response cycle:

- Client (browser) sends an HTTP GET request to a URL
- Server locates the file and returns an HTTP response (status code + HTML)
- Browser parses the HTML, requests linked CSS/JS/images, and renders the page

1.8 URL Anatomy

A URL (Uniform Resource Locator) identifies a resource on the web:

```
https://www.example.com:443/courses/html5?page=1#section-3
```

Part	Value and meaning
`scheme`	https:// — the protocol to use
`domain`	www.example.com — resolved by DNS to an IP
`port`	:443 — default for HTTPS (omitted when default)
`path`	/courses/html5 — file location on the server
`query`	?page=1 — parameters passed to the server
`fragment`	#section-3 — jumps to element with id="section-3"

2. What Is HTML5?

2.1 A Brief History

Version	Year & Milestone
HTML 1.0	1991 — Tim Berners-Lee's original proposal; 18 tags
HTML 2.0	1995 — first formal standard
HTML 3.2 / 4.01	1997–1999 — tables, forms, CSS support added
XHTML 1.0	2000 — strict XML-syntax version of HTML 4
HTML5	2014 W3C Recommendation — living standard (WHATWG)

2.2 The Web Trinity

Modern web pages are built from three complementary technologies:

- HTML5 — Structure: defines what content is and means (headings, paragraphs, images)
- CSS3 — Style: controls appearance (colours, fonts, layout, animations)
- JavaScript — Behaviour: adds interactivity (clicks, data fetching, dynamic updates)

2.3 Key Improvements over HTML4

- Semantic layout tags: <header>, <nav>, <main>, <article>, <section>, <footer>
- Native multimedia: <video> and <audio> without plugins
- Enhanced forms: new input types (email, date, range) with built-in validation
- Canvas API: 2D drawing surface via <canvas>
- Local Storage & Session Storage: client-side data persistence
- Geolocation, Web Workers, WebSockets, and more

3. HTML5 Document Syntax

3.1 The Boilerplate

Every HTML5 document begins with this standard structure:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My First Page</title>
  </head>
  <body>
    <h1>Hello, World!</h1>
  </body>
</html>
```

3.2 Line-by-Line Explanation

Line	Purpose
`<!DOCTYPE html>`	Tells the browser this is an HTML5 document; prevents quirks mode rendering
`<html lang="en">`	Root element; lang attribute declares the document language for screen readers & search engines
`<head>`	Container for metadata (not displayed in the browser window)
`<meta charset="UTF-8">`	Self-closing; declares character encoding; required for non-ASCII characters
`<meta name="viewport">`	Self-closing; ensures correct scaling on mobile devices
`<title>`	Sets the browser tab/window title; important for SEO
`<body>`	All visible page content goes here

3.3 Tag Syntax Rules

Concept	Syntax & Example
Opening tag	<tagname> — starts an element:
Closing tag	</tagname> — ends an element:
Self-closing tag	<tagname> — no content, no closing tag: , <img>, <meta>
Attribute	name="value" inside opening tag: <a href="url">
Comment	<!-- This text is ignored by the browser -->
Indentation	2 spaces per nesting level (convention, not required)
Case	Always write tags in lowercase: not <DIV>

4. Core HTML5 Tags

This section is the core reference. For each tag group, the purpose, syntax, and a working example are provided.

4.1 Document Structure Tags

<!DOCTYPE html> <html> <head> <body>

These are the mandatory skeleton of every HTML5 page. <html> is the root, <head> holds invisible metadata, and <body> holds all visible content.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <link rel="stylesheet" href="style.css">
    <script src="app.js" defer></script>
    <title>Document Title</title>
  </head>
  <body>
    <!-- page content here -->
  </body>
</html>
```

Tag	Purpose
<title>	Tab/window title; max ~60 chars for SEO
<meta>	Metadata: charset, viewport, description, author (self-closing)
<link rel="stylesheet">	Links an external CSS file
<script>	Embeds or links JavaScript; use defer to load after HTML
<style>	Embeds CSS directly in the document

4.2 Semantic Layout Tags (HTML5)

HTML5 replaced generic containers with meaningful tags that describe content roles. These improve accessibility, SEO, and code readability.

```
<body>
  <header>
    <nav><a href="/">Home</a> | <a href="/about">About</a></nav>
  </header>

  <main>
    <section>
      <h2>Latest News</h2>
      <article>
        <h3>HTML5 Is Everywhere</h3>
        <p>Article body text here...</p>
      </article>
    </section>
    <aside>
      <p>Related: CSS3 Tutorial</p>
    </aside>
  </main>

  <footer>
    <p>&copy; 2025 My Website</p>
```

```
  </footer>
</body>
```

Tag	Role & When to Use
<header>	Page banner or section header (logo, nav, headings)
<nav>	Primary navigation menu only — not every link group
<main>	The main unique content of the page; one per document
<section>	A thematic grouping with its own heading
<article>	Self-contained content that makes sense independently (blog post, news item)
<aside>	Tangentially related content: sidebars, pull quotes, ads
<footer>	Closing info: copyright, links, contact details

4.3 Heading Tags

<h1> <h2> <h3> <h4> <h5> <h6>

Headings define document hierarchy. <h1> is the highest level (page title). Always maintain logical order — never skip levels for visual effect.

```
<h1>Web Development Course</h1>
  <h2>Module 1: HTML5</h2>
    <h3>1.1 Document Structure</h3>
    <h3>1.2 Semantic Tags</h3>
  <h2>Module 2: CSS3</h2>
```

Rule: Use exactly one <h1> per page (the page title). Use <h2> for major sections, <h3> for sub-sections.

4.4 Text Content Tags

```
<p>A paragraph of text.</p>

<p>Use <strong>strong</strong> for important text (bold).</p>
<p>Use <em>emphasis</em> for stress (italic).</p>

<div class="container">                  <!-- block-level wrapper -->
  <span style="color:red">Red word</span> in a sentence.
</div>

<p>Line one<br>Line two on the next line.</p>

<hr>   <!-- horizontal rule: thematic break -->

<blockquote cite="https://www.w3.org">
  The Web is for everyone.
</blockquote>

<p><abbr title="HyperText Markup Language">HTML</abbr> is the backbone of the
Web.</p>

<pre>
  function greet() {
    return 'hello';    // whitespace preserved
  }
</pre>
```

```
<p>Install with: <code>npm install html5</code></p>
```

Tag	Purpose & Notes
<p>	Paragraph; block element; browsers add spacing above and below
** **	Generic block container; no semantic meaning; use for layout
****	Generic inline container; use only when no semantic tag fits
****	Important text (rendered bold); has semantic weight unlike CSS bold
****	Emphasised text (rendered italic); has semantic weight
** **	Line break; self-closing; use sparingly — prefer CSS margin
<hr>	Thematic break (rendered as a horizontal line); self-closing
<blockquote>	Long quotation; cite attribute gives the source URL
<q>	Short inline quotation; browser adds quotation marks
<abbr title>	Abbreviation with tooltip showing full term
<pre>	Preformatted text; preserves whitespace and line breaks
<code>	Inline code snippet; rendered in monospace

4.5 List Tags

```
<!-- Unordered list (bullets) -->
<ul>
  <li>HTML5</li>
  <li>CSS3</li>
  <li>JavaScript</li>
</ul>

<!-- Ordered list (numbered) -->
<ol type="1" start="1">
  <li>Download a code editor</li>
  <li>Create index.html</li>
  <li>Open in browser</li>
</ol>

<!-- Definition list -->
<dl>
  <dt>HTML</dt>
    <dd>HyperText Markup Language — the structure of web pages</dd>
  <dt>CSS</dt>
    <dd>Cascading Style Sheets — the appearance of web pages</dd>
</dl>
```

Tag	Purpose
** **	Unordered (bulleted) list; order does not matter
** **	Ordered (numbered) list; type= 1, A, a, I, i; start= first number
** **	List item; used inside or
<dl>	Definition/description list
<dt>	Definition term (the word being defined)
<dd>	Definition description (the explanation, indented)

4.6 Link Tags

```
<!-- Basic hyperlink -->
<a href="https://www.w3.org">Visit W3C</a>

<!-- Open in new tab -->
<a href="https://example.com" target="_blank" rel="noopener noreferrer">
  External Link
</a>

<!-- Relative link -->
<a href="../about/index.html">About Us</a>

<!-- Anchor link (jump to section on same page) -->
<a href="#section-4">Skip to Tags</a>
<h2 id="section-4">Section 4</h2>

<!-- Email and phone links -->
<a href="mailto:info@example.com">Email Us</a>
<a href="tel:+233201234567">Call Us</a>
```

Attribute	Purpose
href	URL of the destination (absolute, relative, anchor, mailto, tel)
target="_blank"	Open link in a new browser tab

rel="noopener"	Security: prevents the new tab from accessing window.opener
title	Tooltip shown on hover
download	Prompt user to download the file instead of navigating to it

4.7 Image & Media Tags

```
<!-- Image: always include alt text -->
<img src="images/chart.png" alt="Bar chart showing 2024 sales data"
     width="800" height="400" loading="lazy">

<!-- Figure with caption -->
<figure>
  <img src="photo.jpg" alt="Students in a computer lab">
  <figcaption>Fig 1. Students working on HTML projects.</figcaption>
</figure>

<!-- Video with multiple formats -->
<video controls width="640" height="360" poster="thumb.jpg">
  <source src="lecture.mp4"  type="video/mp4">
  <source src="lecture.webm" type="video/webm">
  Your browser does not support HTML5 video.
</video>

<!-- Audio -->
<audio controls>
  <source src="podcast.mp3" type="audio/mpeg">
  <source src="podcast.ogg" type="audio/ogg">
</audio>
```

Tag/Attribute	Purpose
<img src>	Embeds an image; self-closing; src = file path
alt	Mandatory: describes image for screen readers and when image fails
width / height	Reserves space; prevents layout shift during load
loading="lazy"	Defers off-screen images until needed (performance boost)
<figure>	Groups image/video with its caption as a semantic unit
<figcaption>	Caption for the <figure> element
<video controls>	Native video player; controls shows play/pause/volume
<audio controls>	Native audio player
<source>	Provides alternative formats for <video>/<audio>; self-closing

4.8 Table Tags

Use tables for tabular data only — never for page layout.

```
<table>
  <thead>
    <tr>
      <th>Student</th>
      <th>Midterm</th>
      <th>Final</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Ama Mensah</td>
      <td>74</td>
      <td>81</td>
    </tr>
    <tr>
      <td colspan="2">Class Average</td>
      <td>78</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td colspan="3">Scores out of 100</td>
    </tr>
  </tfoot>
</table>
```

Tag/Attr	Purpose
<table>	The table container
<thead>	Header row group (sticky/distinct styling)
<tbody>	Main data rows
<tfoot>	Footer row group (totals, notes)
<tr>	Table row
<th>	Header cell; bold & centred by default; use scope= for accessibility
<td>	Data cell
colspan="n"	Cell spans n columns horizontally
rowspan="n"	Cell spans n rows vertically

4.9 Form Tags

Forms collect user input. The action attribute sets the server URL that processes the data; method sets HTTP verb (get or post).

```
<form action="/register" method="post">

  <fieldset>
    <legend>Personal Information</legend>

    <label for="fullname">Full Name *</label>
    <input type="text" id="fullname" name="fullname"
           placeholder="e.g. Ama Mensah" required minlength="2"
maxlength="80">

    <label for="email">Email *</label>
    <input type="email" id="email" name="email" required>
```

```
    <label for="dob">Date of Birth</label>
    <input type="date" id="dob" name="dob" min="1900-01-01">

    <label for="course">Course *</label>
    <select id="course" name="course" required>
      <optgroup label="Computing">
        <option value="">-- Select --</option>
        <option value="cs101">CS101: Intro to Computing</option>
        <option value="web201">WEB201: Web Development</option>
      </optgroup>
    </select>

    <label for="bio">Short Bio</label>
    <textarea id="bio" name="bio" rows="4" cols="50"
              placeholder="Tell us about yourself..."></textarea>

    <label>
      <input type="checkbox" name="terms" required>
      I agree to the Terms &amp; Conditions
    </label>
  </fieldset>

  <button type="submit">Register</button>
  <button type="reset">Clear Form</button>

</form>
```

Input Type	Renders as / Validates
`type="text"`	Single-line text field
`type="email"`	Text field; validates @ and domain format
`type="password"`	Masked text field
`type="number"`	Numeric spinner; min= max= step= attributes
`type="date"`	Date picker; min= max= in YYYY-MM-DD format
`type="checkbox"`	Tick box; value sent only if checked
`type="radio"`	Single-choice buttons within same name= group
`type="file"`	File upload dialog; accept= limits file types
`type="range"`	Slider; min= max= step=
`type="color"`	Colour picker
`type="submit"`	Submit button; triggers form submission
`type="reset"`	Reset button; clears all fields to defaults

Validation Attribute	Purpose
`required`	Field must not be empty
`minlength / maxlength`	Minimum / maximum character count
`min / max`	Minimum / maximum numeric or date value
`pattern="regex"`	Custom validation regex (e.g. [A-Z]{3}\d{3})
`placeholder`	Grey hint text shown when field is empty
`disabled`	Field visible but not interactive; value not submitted

readonly	Value visible and submitted but not editable

4.10 Miscellaneous & HTML5-Specific Tags

```
<!-- Canvas: 2D drawing via JavaScript API -->
<canvas id="myCanvas" width="400" height="200">
  Canvas not supported.
</canvas>

<!-- Details/Summary: native accordion -->
<details>
  <summary>Click to expand: What is HTML?</summary>
  <p>HTML is the standard markup language for creating web pages.</p>
</details>

<!-- Progress bar -->
<progress value="65" max="100">65%</progress>

<!-- Meter: a gauge within a known range -->
<meter value="7.5" min="0" max="10" low="3" high="8"
optimum="9">7.5/10</meter>",

<!-- Semantic time -->
<time datetime="2025-05-01">1 May 2025</time>

<!-- Highlighted text -->
<p>The exam is on <mark>Friday 16 May</mark>.</p>

<!-- Superscript / Subscript -->
<p>E = mc<sup>2</sup>    H<sub>2</sub>O</p>

<!-- Inline frame (embed another page) -->
<iframe src="https://www.openstreetmap.org/export/embed.html"
        width="600" height="400" title="Campus map"></iframe>
```

Tag	Purpose
<canvas>	Blank bitmap surface; draw 2D graphics with JavaScript canvas API
<details>/<summary>	Native show/hide accordion; no JavaScript required
<progress>	Task completion bar (0 to max)
<meter>	Measurement within a range (e.g. score, disk usage)
<time datetime>	Machine-readable date/time; improves SEO and screen readers
<mark>	Highlighted/relevant text (yellow background by default)
<small>	Fine print, copyright notices
<sup>	Superscript (footnotes, exponents)
<sub>	Subscript (chemical formulae)
<iframe>	Embeds another HTML page or external content inline

5. Global Attributes

Global attributes can be applied to any HTML element regardless of type.

Attribute	Value / Example	Purpose
`id`	id="main-nav"	Unique identifier; used by CSS (#id), JS, and anchor links
`class`	class="btn primary"	Space-separated CSS classes; used for styling groups of elements
`style`	style="color:red"	Inline CSS; use sparingly — prefer external stylesheets
`title`	title="Tooltip text"	Advisory tooltip shown on hover
`lang`	lang="fr"	Language of element's content; overrides document lang
`hidden`	hidden	Hides element from display and assistive technology
`tabindex`	tabindex="0"	Controls keyboard tab order; 0 = natural order; -1 = not focusable
`contenteditable`	contenteditable="true"	Makes element editable by the user in-browser
`data-*`	data-user-id="42"	Custom data attributes; accessible via JS dataset property
`draggable`	draggable="true"	Enables HTML5 drag-and-drop API
`spellcheck`	spellcheck="false"	Enables/disables browser spell checking

6. HTML5 Best Practices & Accessibility

6.1 Structural Best Practices

- Always start with <!DOCTYPE html> — prevents browser quirks mode
- Include lang attribute on <html> — required for screen readers
- Set <meta charset="UTF-8"> — handles all Unicode characters
- Use semantic tags (<header>, <main>, <nav>) instead of generic containers
- Maintain a logical heading hierarchy: h1 → h2 → h3 (never skip levels)
- Only one <h1> per page — use it for the page's primary topic

6.2 Accessibility (a11y)

- Every <img> must have a meaningful alt attribute; empty alt="" for decorative images
- Every form <input> must have a corresponding <label for="id">
- Provide captions/transcripts for <video> and <audio>
- Ensure sufficient colour contrast (4.5:1 ratio for normal text, WCAG AA)
- Add title attribute to <iframe> elements
- Use WAI-ARIA roles (role="banner", aria-label, aria-live) when HTML semantics are insufficient

6.3 Validation & Performance

- Validate HTML at https://validator.w3.org — catches missing closing tags, illegal nesting
- Place <script> tags at end of <body> or use defer/async to avoid blocking rendering
- Use loading="lazy" on images below the fold
- Keep HTML focused on structure; move all styling to external CSS files

7. Quick Reference: HTML5 Tag Summary

The following table covers all major tags from Section 4. Use it as a daily reference card.

Tag	Category	Self-closing?	Key Attributes
`<!DOCTYPE>`	Document	Yes	—
`<html>`	Document	No	lang
`<head>`	Document	No	—
`<body>`	Document	No	—
`<title>`	Document	No	—
`<meta>`	Document	Yes	charset, name, content, viewport
`<link>`	Document	Yes	rel, href, type
`<script>`	Document	No	src, defer, async, type
`<style>`	Document	No	—
`<header>`	Semantic	No	—
`<nav>`	Semantic	No	—
`<main>`	Semantic	No	—
`<section>`	Semantic	No	—
`<article>`	Semantic	No	—
`<aside>`	Semantic	No	—
`<footer>`	Semantic	No	—
`<h1>–<h6>`	Heading	No	—
`<p>`	Text	No	—
`<div>`	Text	No	id, class
`<span>`	Text	No	id, class
`<strong>`	Text	No	—
`<em>`	Text	No	—
` `	Text	Yes	—
`<hr>`	Text	Yes	—
`<blockquote>`	Text	No	cite
`<pre>/<code>`	Text	No	—
`<abbr>`	Text	No	title
`<ul>/<ol>`	List	No	type, start (ol only)
`<li>`	List	No	value
`<dl>/<dt>/<dd>`	List	No	—
`<a>`	Link	No	href, target, rel, download
`<img>`	Media	Yes	src, alt, width, height, loading
`<figure>`	Media	No	—
`<figcaption>`	Media	No	—
`<video>`	Media	No	src, controls, autoplay, loop, muted, poster, width
`<audio>`	Media	No	src, controls, autoplay, loop

Tag	Category	Self-closing?	Key Attributes
`<source>`	Media	Yes	src, type
`<table>`	Table	No	—
`<thead>/<tbody>/<tfoot>`	Table	No	—
`<tr>`	Table	No	—
`<th>/<td>`	Table	No	colspan, rowspan, scope
`<form>`	Form	No	action, method, enctype
`<input>`	Form	Yes	type, id, name, value, required, placeholder
`<label>`	Form	No	for
`<textarea>`	Form	No	rows, cols, name, placeholder
`<select>`	Form	No	name, multiple, required
`<option>`	Form	No	value, selected, disabled
`<button>`	Form	No	type (submit/reset/button)
`<fieldset>`	Form	No	—
`<legend>`	Form	No	—
`<canvas>`	HTML5	No	id, width, height
`<details>`	HTML5	No	open
`<summary>`	HTML5	No	—
`<progress>`	HTML5	No	value, max
`<meter>`	HTML5	No	value, min, max, low, high, optimum
`<time>`	HTML5	No	datetime
`<mark>`	HTML5	No	—
`<iframe>`	Embed	No	src, title, width, height, loading

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